

Akshay Dahiya

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Experience

Software Engineer

April 2020 - Present

HYPERSOX INC.

- Designed API Gateway and Management Platform from scratch for better performance and speedy development.
- Improved API performance by about 80% by optimizing the codebase and incorporating compression algorithms like Brotli
- Redesigned the entire authentication mechanism using Auth0 to make it more secure and robust.
- Designed multiple critical components like the ELK logging and performance monitoring system, Learning Center, etc.

Master's Project & Special Problems

Jan 2020 - May 2021

GEORGIA INSTITUTE OF TECHNOLOGY

- Working under **Dr. Thad Starner** to develop tools to Counter Plagiarism and assist online Education at Scale.
- Our research project **Jack Watson** got featured in **Forbes**, **TimesHigherEducation**, and multiple other places.
- Worked on 4 research projects to Counter Plagiarism and assist Education at Scale.

Graduate Programmer

June 2018 - Dec 2018

FIDELITY INTERNATIONAL

- Developed a real-time feedback application using web sockets, Django, Redis, and D3.js.
- Worked in the behavioral finance team and helped in designing tools for portfolio managers using React.

Founder

Aug 2015 - Dec 2017

TYPINGEEK

- Founded Typingeeek, a platform to learn to type without wasting time on meaningless phrases.
- Handled everything from User research to product development to final deployment.

Google Summer of Code Participant

May 2017 - Aug 2017

PYTHON SOFTWARE FOUNDATION

- Designed **Hydrus**, a web framework built in Flask that can create semantic web-based Restful APIs using a single specs file (Hydra Vocabulary)
- Build a simulation (**Flock demo**) where multiple servers and clients interact autonomously to demonstrate Semantic Web capabilities.
- The project has 168 Stars and 133 Forks (to date) on Github.

Volunteer Experience

Administrator, Google Summer of Code

Feb 2018 - Present

HYDRAECOSYSTEM.ORG

- Hydra** is a documentation framework based on Linked Data (Semantic Web) tools to build the next generation of connected Web APIs
- We develop tools to make the Hydra framework operational using Python and demonstrate the semantic web capabilities.

Mentor & Project Reviewer

Feb 2017 - Mar 2020

UDACITY

- Assisted **1000+** students with questions related to projects, content and Code for Full Stack and Deep Learning Nanodegrees
- Reviewed **5000+** projects providing valuable insights to students related to better coding and design practices.

Skills

Languages: Python, Javascript, C++(basic)

Web: Django, Flask, FastAPI, React, PostgreSQL, Redis, ELK, Auth0, JSON-LD, Celery, Web Sockets, Semantic Web

AI: Numpy, Pandas, Sklearn, OpenCV, Tensorflow, Pytorch, Fast.ai

Other: Algorithms, Data Structures, Docker, AWS, GCP, Git, Linux, Inkscape, UX Research, UI Design, Unit testing, Web scraping

Education

Georgia Institute of Technology, Atlanta, USA

Jan 2019 - May 2021

MASTER OF SCIENCE IN COMPUTER SCIENCE

- GPA 4.0/4.0 (12 CS research credits)
- Machine Learning Specialization

DCRUST, Murthal, India

July 2014 - Apr 2018

B.TECH IN ELECTRONICS AND COMMUNICATION

- CGPA 7.05/10

Publications

1. Akshay Dahiya, Rocko Graziano, India Irish, and Thad Starner. 2021. JackMarker with GitDown: A Framework to Counter Plagiarism at Scale. In Proceedings of the Eighth ACM Conference on Learning @ Scale (L@S'21). Association for Computing Machinery, New York, NY, USA, 335–338. DOI: <https://doi.org/10.1145/3430895.3460982>

2. Raghav Apoorv, Akshay Dahiya, Uma Sreeram, Bharat Rahuldhev Patil, India Irish, Rocko Graziano, and Thad Starner. 2020. Examiner: A Plagiarism Detection Tool for Take-Home Exams. In Proceedings of the Seventh ACM Conference on Learning @ Scale (L@S '20). Association for Computing Machinery, New York, NY, USA, 261–264. DOI: <http://dx.doi.org/10.1145/3386527.3406723>